



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2024

ME1103-1 – THEORY OF MACHINES
(ME)

Duration: 3 Hours

Max. Marks: 100

Note:

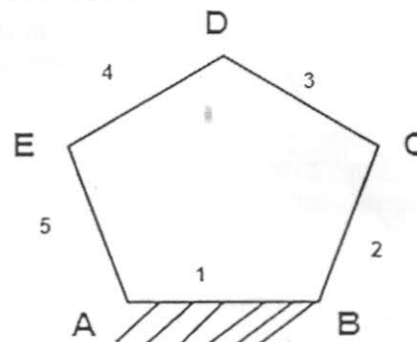
Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- If the relative motions of the rigid bodies are in one plane or in parallel planes, then the mechanism is called
 A) spherical mechanism
 B) planar mechanism
 C) spatial mechanism
 D) flexure mechanism
- According to Kutzbach criterion, the mobility of a rigid body is given by the formula:
 A) $3(n+1)-2j_1-j_2$
 B) $3(n+1)+2j_1+j_2$
 C) $3(n-1)+2j_1+j_2$
 D) $3(n-1)-2j_1-j_2$
- Find the mobility of the given mechanism.



- Find the mobility of the given mechanism.
 A) 0
 B) 1
 C) 2
 D) 3
- If the link next to the shortest link is fixed, what type of mechanism is obtained?
 A) Double crank mechanism
 B) Linkage is not planar
 C) Double rocker mechanism
 D) **Crank rocker mechanism**
- Often an unbalance of forces is produced in rotary or reciprocating machinery due to the
 A) Centripetal forces
 B) Centrifugal forces
 C) Friction forces
 D) Inertia forces
- In a revolving rotor, the centrifugal force remains balanced as long as the centre of the mass of the rotor lies-
 A) Below the axis of shaft
 B) On the axis of the shaft
 C) Above the axis of shaft
 D) Away from the axis of shaft
- Which of the following is true for centrifugal force causing unbalance?
 A) Direction changes with rotation
 B) Magnitude changes with rotation
 C) Direction and magnitude both change with rotation
 D) Direction and magnitude both remain unchanged with rotation
- If the unbalanced system is not set right, then
 A) Static forces develop
 B) Dynamic forces develop
 C) Tangential forces develop
 D) Radial forces develop

9. The size of a cam depends upon
 A) base circle
 C) pitch circle
 B) pitch curve
 D) prime circle
10. The angle between the direction of the follower motion and a normal to the pitch curve is called
 A) Base angle
 C) Prime angle
 B) Pressure angle
 D) Pitch angle
11. In a radial cam, the follower moves
 A) along the cam axis
 C) in a direction perpendicular to the cam axis
 B) in a direction parallel to the cam axis
 D) in any direction irrespective of the cam axis
12. Which of the following statements is true for cam profile?
 A) Pitch point on the pitch curve has minimum pressure angle
 C) Pitch circle is drawn through trace point from the center of cam
 B) In case of roller follower, trace point represents centre of the roller
 D) Base circle is the largest circle drawn from the center of cam
13. A disc is spinning with an angular velocity ω rad/s about the axis of spin. The couple applied to the disc causing precession will be
 A) $1/2 I \omega^2$
 C) $1/2 I \omega \omega_p$
 B) $I \omega^2$
 D) $I \omega \omega_p$
14. A disc spinning on its axis at 20 rad/sec will undergo precession when a torque of 100 N-m is applied about an axis normal to it. If the mass moment of inertia of the disc is 1 kg-m^2 . The precessional Velocity is
 A) 2 rad/s
 C) 10 rad/s
 B) 5 rad/s
 D) 20 rad/s
15. The engine of an aeroplane rotates in clockwise direction when seen from the tail end and the aeroplane takes a turn to the left. The effect of the gyroscopic couple on the aeroplane will be
 A) to raise the nose and dip the tail
 C) to raise the nose and tail
 B) to dip the nose and raise the tail
 D) to dip the nose and tail
16. In an automobile moving in forward direction, if the vehicle makes a left turn, the gyroscopic torque
 A) increases the forces on the outer wheels
 C) Increases the forces in the front axle wheels
 B) decreases the forces on the outer wheels
 D) Decreases the forces on the rear axle wheels
17. The smaller and generally the driving gear of a pair of mating gears is called _____
 A) pitch
 C) rack
 B) module
 D) pinion
18. The number of teeth per unit length of the pitch circle diameter is called
 A) diametral pitch
 C) clearance
 B) Module
 D) gear ratio
19. Fluid resistance causes damping which is known as _____
 A) Resistance damping
 C) Viscous damping
 B) Fluid damping
 D) Liquid damping
20. In damped vibrations, the amplitude of the resulting vibration gradually reduces. This is due to the reason that an amount of energy is always dissipated to overcome the
 A) Frictional resistance
 C) Fluid pressure
 B) Work done
 D) Air pressure

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) Determine the mobility of the system shown in Figure 1.

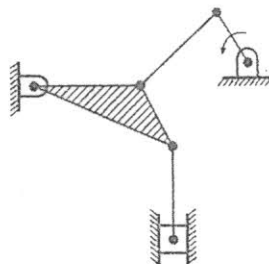


Figure1

4 L*2 01 01

- b) What are ternary and quaternary links?
- c) For the static equilibrium of the mechanism shown in Figure 2, determine the torque to be applied on link AB. Link lengths are as follows AB=50mm, BC=66mm, CD= 55mm, AD=100mm, CE=25mm.

4 L1 01 01

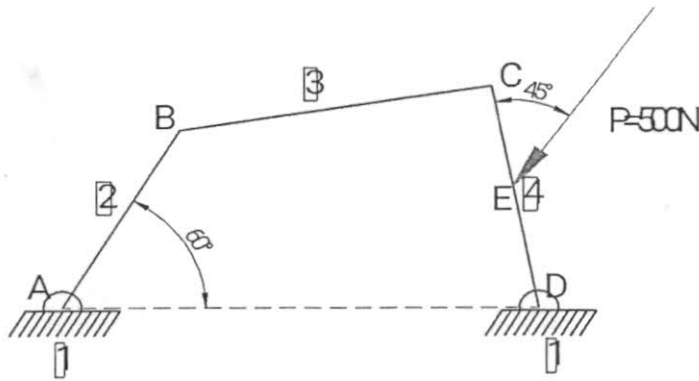


Figure 2

8 L3 02 02

2. a) For the static equilibrium of the mechanism shown in figure 3, Determine the magnitude of force Q to be applied on link 2. Link lengths are as follows: $O_2A=60\text{mm}$, $BD=40\text{mm}$, $AB=120\text{mm}$, $O_2E=30\text{mm}$.

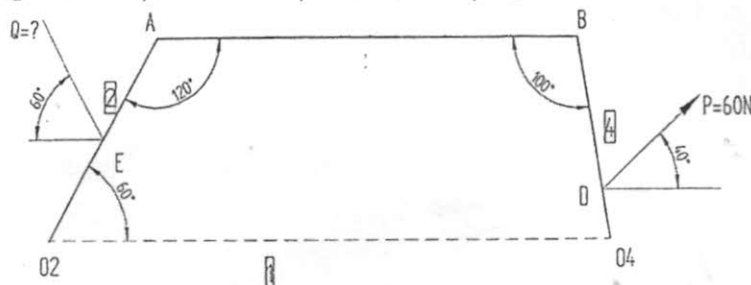


Figure 3

8 L3 02 02

- b) Explain various inversions of single slider crank mechanism.
3. a) What is meant by Quick Return Mechanism? Sketch and explain Crank and slotted lever mechanism.
- b) Define Static and Dynamic balancing.
- c) List out the steps involved in the graphical solution of balancing of several masses rotating in same plane.

8 L2 01 02

8 L2 01 01

4 L1 02 01

4 L2 02 01

Unit - II

4. a) What is cam and follower? With a neat sketch explain how cams are classified.
- b) Derive an expression for stability of a 4 wheel automobile.
5. a) Design a cam for operating the exhaust valve of an oil engine. It is required to give equal uniform acceleration and retardation during opening and closing of the valve each of which corresponds to 60° of cam rotation. The valve must remain in the fully open position for 20° of cam rotation. The lift of the valve is 37.5 mm and the least radius of the cam is 40 mm. The follower is provided with a roller of radius 20 mm and its line of stroke passes through the axis of the cam.
- b) A 4-wheel automobile travelling along a track of 100m of mean radius, each of the four wheels have moment of inertia of 2.5kgm^2 and effective radius of 0.3m. The rotating parts of the engine have MI of 1.2kgm^2 . Assume the engine axis is parallel to the front axle and the crankshaft rotates in the same direction as that of the road wheels. Ratio of the engine speed to the speed of the wheels is 3:1. The automobile has a mass of 1600kg, and its centre of gravity is 0.5m above the ground level width of the track is 1.5m. Determine the limiting speed of the vehicle around the curve for all the 4 wheels to maintain contact with the road surface. Assume CG of the vehicle lies at the centre.

8 L1 03 01

8 L2 03 01

8 L3 03 02

8 L3 03 02

6. a) Derive an expression for displacement, velocity, acceleration of a follower executing simple harmonic motion. Also, plot the variation. 8 L2 03 01
- b) A four-wheel motor car of mass 2 tons has height of CG 60cm above the ground level. Engine parts and transmission are equivalent to a mass of 80kg with radius of gyration of 15cm. Axis of the engine and wheel coincide with each other. The car negotiates a curve of 60m radius at 72km/hr with overall gear ratio of 4:1. Radius of the road wheel 30cm and I or Moment of Inertia is 3kgm^2 . Assuming width of the track as 1.5m. Determine reaction on each wheel. 8 L3 03 02

Unit – III

7. a) In a reverted epicyclic gear train, the arm A carries two gears B and C and a compound gear D - E. The gear B meshes with gear E and the gear C meshes with gear D. The number of teeth on gears B, C and D are 75, 30 and 90 respectively. Find the speed and direction of gear C when gear B is fixed and the arm A makes 100 r.p.m. clockwise. 8 L3 04 02
- b) Determine the natural frequency of the compound pendulum. 8 L2 05 01
8. a) State and prove Law of gearing. Obtain the expression for velocity of sliding. 8 L2 04 01
- b) A circular cylinder of mass m and radius r is connected by a spring of stiffness K as shown in Figure 8b. If it is free to roll on the rough surface which is horizontal without slipping, find its natural frequency.

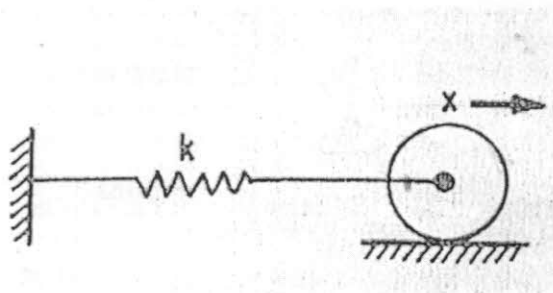


Figure 8b.

8 L3 05 02

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
